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3031ICT: Applied Data Mining
3804ICT: Data Mining

Lecture 11: Data Mining on Information Networks



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Data Mining on Information Networks

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- Information Networks
  - Social Network Analysis
  - Centrality
  - Prestige
- Sentiment Analysis
  - SA Problems
  - Document sentiment classification
  - Sentence subjectivity & sentiment classification
- Summary

Information Networks

- **Information networks**: consist of a large number of interacting, multi-typed components, such as **human social activities**, **communication** and **computer systems**, as well as **biological networks**.
- The information network analysis has gained extremely wide attentions from researchers in many disciplines, such as **computer science**, **social science**, **physics**, and so on.
- The basic paradigm is to mine hidden patterns through mining link relations from **networked** data.

Social Network Analysis

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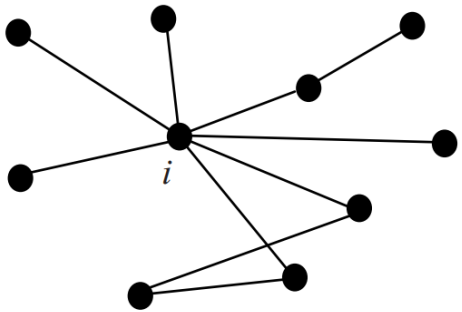
- Social network is the study of social entities (people in an organization, called **actors**), and their interactions and relationships.
- The interactions and relationships can be represented with a network or graph,
  - each vertex (or node) represents an actor and
  - each link represents a relationship
- From the network, we can study the properties of its structure, and the role, position and prestige of each social actor.
- We can also find various kinds of sub-graphs, e.g., **communities** formed by groups of actors.

Social Network and Web

- Social network analysis is useful for the Web
  - Each page: a social actor and
  - Each hyperlink: a relationship
- Many results from social network can be adapted and extended for use in the Web context.
- We study two types of social network analysis, **centrality** and **prestige**, which are closely related to hyperlink analysis and search on the Web.

Centrality

- Important or prominent actors are those that are linked or involved with other actors extensively.
- A person with extensive contacts (links) or communications with many other people in the organization is considered more important than a person with relatively fewer contacts.
- The links can also be called **ties**. A central actor is one involved in many ties.

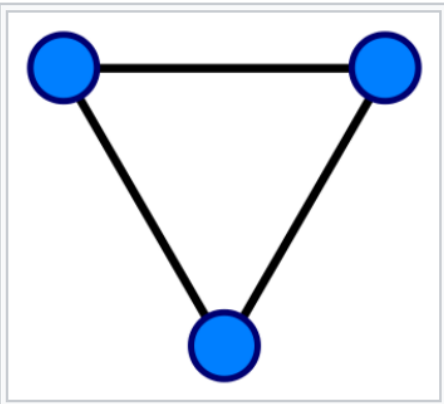


Degree Centrality

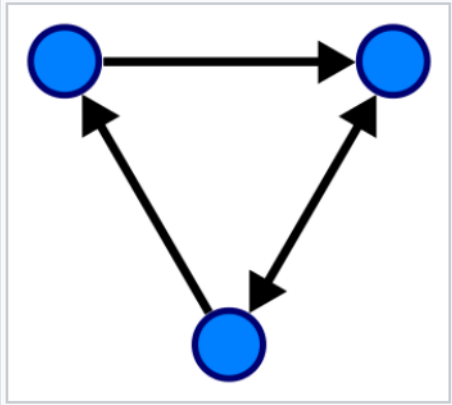
- **Undirected Graph:** In an undirected graph, the **degree centrality** of an actor *i* (denoted by $C_D(i)$) is simply the node degree (the number of edges) of the actor node, denoted by $d(i)$, normalized with the maximum degree, n-1.

$$C_D(i) = \frac{d(i)}{n-1}.$$

- The value of this measure ranges between 0 and 1 as n-1 is the maximum value of $d(i)$.



A graph with three vertices and three edges.



A directed graph with three vertices and four directed edges (the double arrow represents an edge in each direction).

Closeness Centrality

- The basic idea of closeness centrality is that an **actor** i is central if it **can easily interact with all other actors**.
- Let the shortest distance from actor i to actor j be d(i, j) (measured as the number of links in a shortest path).
- **Undirected Graph**: The closeness centrality $C_C(i)$ of actor i is defined as

$$C_C(i) = \frac{n-1}{\sum_{j=1}^{n} d(i,j)}.$$

- The value of this measure also ranges between 0 and 1 as *n*-1 is the minimum value of the denominator, which is the sum of the shortest distances from i to all other actors. Note that this equation is only meaningful for a connected graph.

Prestige

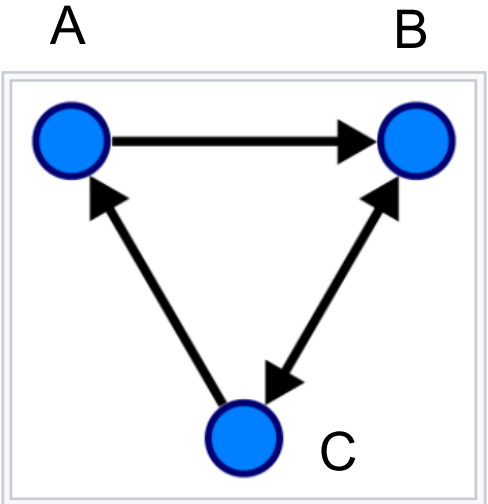
- Prestige is a more refined measure of prominence of an actor than centrality.
  - Distinguish: ties sent (out-links) and ties received (in-links).
- A prestigious actor is one who is the object of extensive ties as a recipient.
  - To compute the prestige: we use only in-links.
- Difference between centrality and prestige:
  - centrality focuses on out-links
  - prestige focuses on in-links.
- We study three prestige measures. **Rank prestige** forms the basis of most Web page link analysis algorithms, including PageRank and HITS.

Degree Prestige

- The simplest measure of prestige of an actor i is its in-degree:

$$P_D(i) = \frac{d_I(i)}{n-1}$$

- where $d_I(i)$ is the in-degree of i (the number of in-links of i) and n is the total number of actors in the network.
- In the degree prestige, dividing by $n - 1$ standardizes the prestige value to the range from 0 and 1. The maximum prestige value is 1 when every other actor links to, or chooses actor i.



A directed graph with three vertices and four directed edges (the double arrow represents an edge in each direction).

Proximity Prestige

- The degree index of prestige of an actor *i* only considers the actors that are adjacent to *i*.
- The **proximity prestige** generalizes it by considering both the actors directly and indirectly linked to actor *i*.
  - So we consider every actor *j* that can reach *i*.
- Let I_i be the set of actors that can reach actor i, called **influence domain.**
- The **proximity** is defined as closeness or distance of other actors to *i*.
- Let *d*(*j, i*) denote the shortest path distance from actor *j* to actor *i*.
- Each link has the unit distance. The average distance:

$$\frac{\sum_{j \in I_i} d(j,i)}{|I_i|}$$

Proximity prestige (cont …)

- The proximity prestige is

$$P_P(i) = \frac{|I_i|/(n-1)}{\sum_{j \in I_i} d(j,i) \Big/ |I_i|}$$

- The numerator is the proportion of actors that can reach actor i.
- The denominator is the average distance.

Rank Prestige

- In the previous two prestige measures, an important factor is not considered,
  - the **prominence** of individual actors who do the "voting"
- In the real world, a person *i* chosen by an important person is more prestigious than chosen by a less important person.
  - For example, if a company CEO votes for a person, this is much more important than a worker votes for the person.
- If one's circle of influence is full of prestigious actors, then one's own prestige is also high.
  - Thus one's prestige is affected by the rank or status of the involved actors.

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Introduction to Sentiment Analysis

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- Sentiment analysis (SA) or opinion mining
  - computational study of opinion, sentiment, appraisal, evaluation, and emotion.
- Why is it important?
  - Opinions are key influencers of our behaviors.
    - Rise of social media –> opinion data
  - Rise of AI and chatbots:
    - Emotion and sentiment are key to human communication

Terms defined - Merriam-Webster

- *Sentiment*: an attitude, thought, or judgment prompted by feeling.
  - A sentiment is more of a feeling.
  - "*I am concerned about the current state of the economy.*"
- *Opinion:* a view, judgment, or appraisal formed in the mind about a particular matter.
  - a concrete view of a person about something.
  - "*I think the economy is not doing well.*"

SA: A fascinating problem!

- Intellectually challenging & many applications
  - A popular research area in NLP, and data mining
  - spread from CS to management and social sciences
  - A large number of companies in the space globally
    - > 300 in the US alone.
- It touches every aspect of NLP & also is confined.
  - A “simple” semantic analysis problem.
- A major technology from NLP.
  - But it is hard.

Two main types of opinions

- Regular opinions: Sentiment/opinion expressions **on some target entities**
  - Direct opinions:
    - "The touch screen is really cool."
  - Indirect opinions:
    - "After taking the drug, my pain has gone."
- Comparative opinions: Comparison of **more than one entity**
  - E.g., "iPhone is better than Blackberry."
- We focus on regular opinions and just call them opinions.

Definition of Opinion

- **ID: abc123 on 29-09-2026 --** " (1) *I bought an iPhone 18 Pro yesterday.* (2) *It is such a nice phone.* (3) *The touch screen is really* ***cool****.* (4) *The voice quality is* ***great*** *too. It is much* ***better*** *than my Samsung S25.* (5) *However, my mum was* ***mad*** *with me as I didn't tell her before I bought the phone.* (6) *She thought the phone was too* ***expensive***"

- **Definition**: An *opinion* is a tuple: (*target*, *sentiment*, *holder*, *time*)

- This definition is concise, but not easy to use.
  - Target can be complex, e.g.,"*I bought an iPhone. The voice quality is amazing.*"
    - *Target* = *voice quality*?

A more practical definition

- An *opinion* is a quintuple

  (*entity*, *aspect*, *sentiment*, *holder*, *time*)

  where

  - ***entity****:* target entity (or object).
  - ***Aspect***: aspect (or feature) of the entity.
  - ***Sentiment***: +, -, or neu, a rating, or an emotion.
  - ***holder***: opinion holder.
  - ***time***: time when the opinion was expressed.

Our example blog in quintuples

- **ID: abc123 on 29-09-2026** "*I bought an iPhone 18 Pro a few days ago. It is such a nice phone. The touch screen is really cool. The voice quality is great too. It is much better than my old Samsung S25, which was a terrible phone and so difficult to type with its tiny keys. However, my mother was mad with me as I did not tell her before I bought the phone. She also thought the phone was too expensive, …*"
- In quintuples

  (iPhone, GENERAL, +, abc123, 29-09-2026)

  (iPhone, touch_screen, +, abc123, 29-09-2026)

  ....

Opinion Summary

- With a lot of opinions, a summary is necessary.
  - Not traditional text summary: from long to short.
  - Text summarization: defined operationally based on algorithms that perform the task
- Opinion summary (OS) can be defined precisely,
  - not dependent on how summary is generated.
- Opinion summary needs to be quantitative
  - 60% positive is very different from 90% positive.
- Main form of OS: ***Aspect-based opinion summary***

Opinion Summary

A common form of summary is based on aspects: Aspect/feature Based Summary of opinions about iPhone

Aspect: **Touch screen**

Positive:

- *The touch screen was really cool.*
- *The touch screen was so easy to use and can do amazing things.*

…

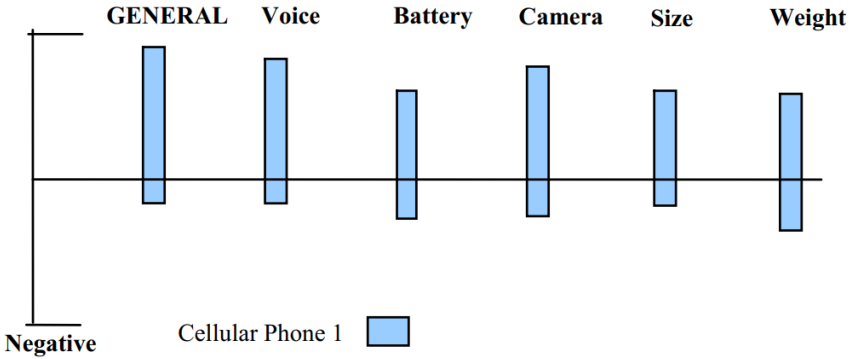
Negative:

- The screen is easily scratched.
- I have a lot of difficulty in removing finger marks from the touch screen.

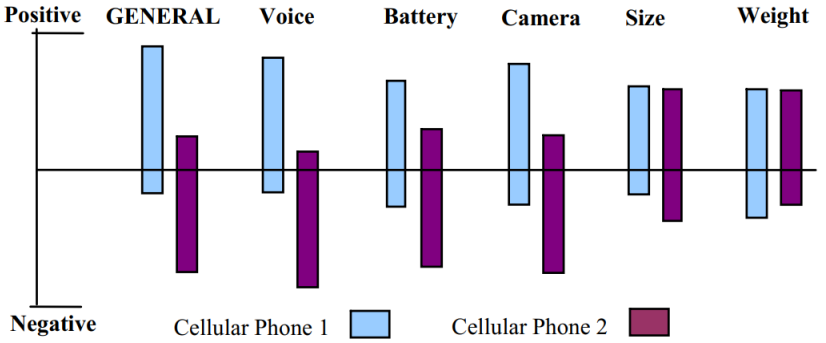
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Aspect: **voice quality**

…



Visualization of aspect-based summary of opinions on a cellular phone



Visual opinion comparison of two cellular phones

Subjective vs Objective

- **Sentence subjectivity**
  - ***An objective sentence*** presents some factual information,
  - while ***a subjective sentence*** expresses some personal feelings, views, emotions, or beliefs.

Subjective Opinions

- **Definition**: A subjective opinion is a regular or comparative opinion given in a subjective statement.
- For example,
  - “Coke tastes great.”
  - “I think Google’s profit will go up next month.”
  - “This camera is a masterpiece.”
  - “We are seriously concerned about this new policy.”
  - “Coke tastes better than Pepsi.”

Objective Sentences with Opinion

- Most opinion sentences are subjective, but objective (or factual) sentences can imply opinions too.
- They express **desirable or undesirable facts**
  - "The machine stopped working in the second day"
  - "We bought the mattress yesterday, and a body impression has formed."
  - "After taking the drug, there is no more pain"
- Such sentences are very hard to handle

Emotion

- Emotion is what SA wants to capture
- No agreed set of basic emotions of people
  - Based on Parrott (2001), people have six basic emotions
    - **love, joy, surprise, anger, sadness, and fear**
- Although related, emotions and opinions are not equivalent
  - Opinion: rational (+/-) view on something
    - Cannot say “I like”
  - Emotion: focusing on an inner feeling
    - Can say “I am angry” or “There is sadness in her eyes”

Definition of Emotion

- **Definition (Emotion)**: It is a quintuple,

  (*entity*, *aspect*, *emotion_type*, *feeler*, *time*)

  - E.g., "*I am so mad with the hotel manager because he refused to refund my booking fee*"
    - Entity: hotel
    - Aspect: manager
    - emotion_type: anger
    - feeler: I
    - time: unknown
  - The definition can also include *the cause*.

Emotion Expressions

- 1. use emotion or mood words or phrases
  - E.g., love, disgust, angry, and upset
- 2. describe emotion-related behaviors,
  - “He cried after he saw his mother” and
  - “After he received the news, he jumped up and down for a few minutes like a small boy.”
- 3. use intensifiers: Common English intensifiers include
  - very, so, extremely, dreadfully, really, awfully, etc.

Emotion Expressions (conted.)

- 4. use superlatives
  - many superlative expressions also express emotions, for example, "*This car is simply the best*"
- 5. use swearing, cursing, insulting, blaming, accusing, and threatening expressions

Assumption and Goal

- Assumption: The doc is written by a single person and express opinion/sentiment on a single entity.
- Reviews usually satisfy the assumption.
  - Almost all research papers use reviews
  - Positive: 4 or 5 stars, negative: 1 or 2 stars
- Forum postings and blogs do not follow the assumption
  - They may mention and compare multiple entities
  - Many such postings express no sentiments

Sentiment Classification

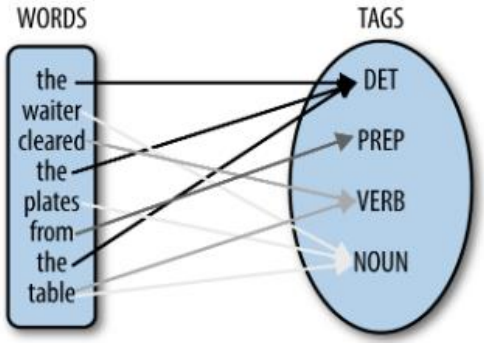
- Classify a whole opinion document (e.g., a review) based on the overall sentiment of the opinion holder
  - Classes: Positive, Negative (possibly Neutral)
- An example review:
  - *"I bought an iPhone a few days ago. It is such a nice phone, although a little large. The touch screen is cool. The voice quality is great too. I simply love it!"*
  - Classification: positive or negative?
- It is basically a text classification problem

Supervised Learning

- Directly apply supervised learning techniques to classify reviews into positive and negative.
- Three classification techniques can be tried:
  - Naïve Bayes, Maximum Entropy, Support Vector Machines (SVM)
- Features: negation tag, unigram (single words), bigram, POS tag, position.
- SVM did the best based on movie reviews.
- Nowadays, deep Learning models with word embedding is state-of-the-art.

Features for Supervised Learning

- The problem has been studied by numerous researchers.
- Key: feature engineering. A large set of features have been tried by researchers. E.g.,
  - Terms frequency and different IR weighting schemes
  - Part of speech (POS) tags
    http://universaldependencies.org/u/pos/all.html#al-u-pos/DET
  - Etc.



Lexicon-based Approach

- Using a set of sentiment terms, called the sentiment lexicon
  - Positive words: great, beautiful, amazing, …
  - Negative words: bad, terrible awful, unreliable, …
- The value for each sentiment term is assigned from [−5, +5].
  - Consider *negation*, *intensifier* (e.g., very), and *diminisher* (e.g., barely)
- Decide the sentiment of a review by aggregating scores from all sentiment terms

Deep Learning

- Recently, deep neural networks have been used for sentiment classification. E.g.,
  - Socher et al used deep learning to work on the sentence parse tree based on words/phrases compositionality in the framework of distributional semantics
  - Many papers …
  - Also related
    - extract opinion expressions
    - identify opinion & target relations

Review Rating Prediction

- Apart from classification of positive or negative sentiments,
  - Research has also been done to predict the rating scores (e.g., 1–5 stars) of reviews.
  - Training and testing are reviews with star ratings.
- Formulation: The problem is formulated as regression since the rating scores are ordinal.

Sentence Sentiment Analysis

- Usually consist of two steps
  - Subjectivity classification (Wiebe et al 1999)
    - To identify subjective sentences
      - Sentence similarity.
      - Naïve Bayesian classification
  - Sentiment classification of subjective sentences
    - Into two classes, positive and negative
- But bear in mind
  - Many objective sentences can imply sentiments
  - Many subjective sentences do not express positive or negative sentiments/opinions

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Have a question?

